

Unit 3

Analysis of Time Series

The time series is arrangement of statistical data in a chronological order and series relating prices, production and consumption of various commodities

$$y = f(t)$$

where y is the value of phenomenon and $f(t)$ is under the consideration of time t

eg: Population of the country in different years
Temperature of different days.

Components of Time Series

1. Secular trend or long term movement
2. Periodic movement or short term fluctuations
 - i) Seasonal variations
 - ii) Cyclic variations
 - iii) Random or irregular variations



linear



non-linear

Secular trend

Increase or decrease or stagnant during long period of time

It is observed in series relating to economics and business in the upward tendency

eg: Production and sales etc

Downward Tendency

eg: The no. of deaths

Methods of measuring trends

- 1) Graphical method
- 2) Method of semi average
- 3) Method of curve fitting by the principle of least square
- 4) Method of moving average.

Semi average

The whole series is divided into 2 equal parts for odd number of years, 2 equal parts are obtained on omitting the value for the middle n years.

1. Medium sized company recorded its annual sales over 6 years period as follows

Year	1982	1983	1984	1985	1986	1987
Sales	10	12	11	16	15	20

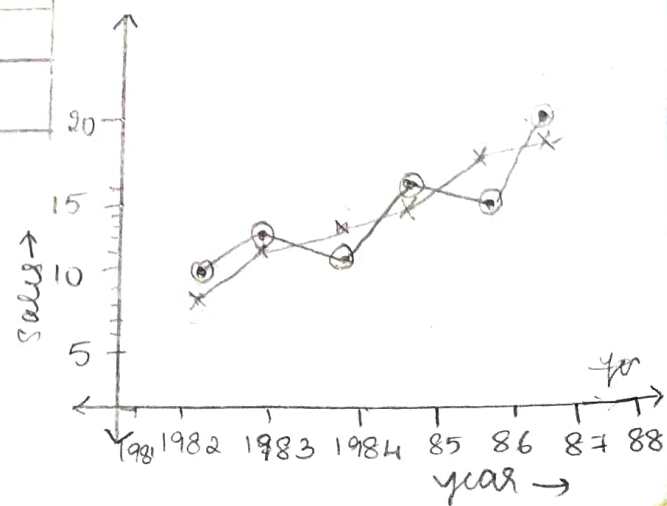
The company ^{management} wants to analyze past performance to improve future sales strategies as a part of analysis team.

Year	Sales		Trend val
1982	10	$\bar{x}_1 = \frac{10+12+11}{3}$ $= 11$	$11 - 2 = 9$
1983	12		Center = 11
1984	11		$11 + 2 = 13$
1985	16	$\bar{x}_2 = \frac{16+15+20}{3}$ $= 17$	$17 - 2 = 15$
1986	15		Center = 17
1987	20		$17 + 2 = 19$

$$\text{annual sales} = \frac{\bar{x}_2 - \bar{x}_1}{3} = \frac{17 - 11}{3} = 2$$

$$\text{for } 1988 \rightarrow 19 + 2 = 21$$

for sales values one line graph & trend value

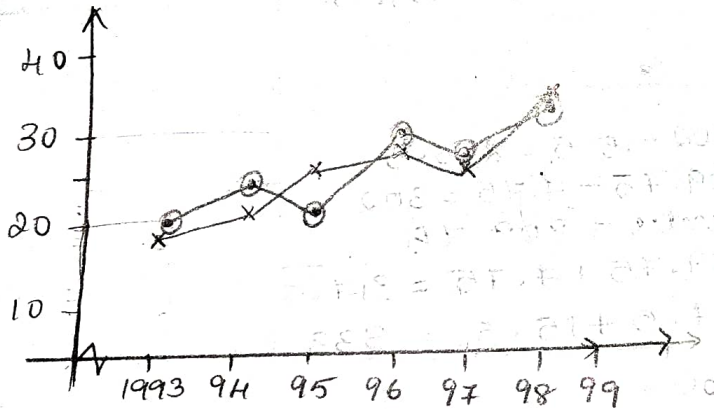


2. Apply the method of semi average to determine the trend of the following data. Estimate the value for the year 2000.

year	sales		Trend val
1993	20	$\bar{x}_1 = \frac{20+24+22}{3} = 22$	$22 - 2.66 = 19.34$
1994	24		center = 22
1995	22		$22 + 2.66 = 24.66$
1996	30	$\bar{x}_2 = \frac{30+28+32}{3} = 30$	$30 - 2.66 = 27.34$
1997	28		center = 30
1998	32		$30 + 2.66 = 32.66$
1999	3		

$$\text{annual sales} = \frac{\bar{x}_2 - \bar{x}_1}{3} = \frac{30 - 22}{3} = 2.66$$

for 2000 the sales is $32.66 + 2.66 + 2.66 = 37.98$



3. Determine the trend using semi average and estimate the sales for 1984

year	sales		Trend val
1975	18	$\bar{x}_1 = \frac{18+24+26+28}{4} = 24$	$22.2 - 3.6 = 18.6$
1976	24		$24 - 1.8 = 22.2$
1977	26		center = 24
1978	28		$24 + 1.8 = 25.8$
1979	33		$25.8 + 3.6 = 29.4$
1980	36	$\bar{x}_2 = \frac{36+40+44+48}{4} = 42$	$42.2 - 3.6 = 38.6$
1981	40		$42 - 1.8 = 40.2$
1982	44		center = 42
1983	48		$42 + 1.8 = 43.8$
			$43.8 + 3.6 = 47.4$

$$\text{annual sales} = \frac{\bar{x}_2 - \bar{x}_1}{5} = \frac{42 - 24}{5} = 3.6$$

$$\text{half yearly increment} = \frac{3.6}{2} = 1.8$$

for 1984 sales is $47.4 + 3.6 = 51$

Draw Graph in test & Exams.

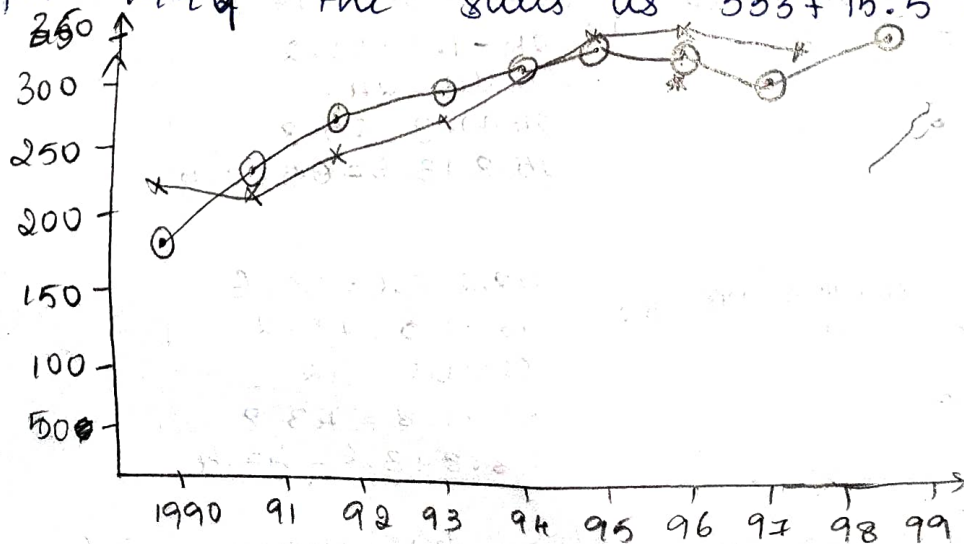
4. From the following series of annual data find trend line by the method of semi average. also find the value of 1999.

Year	Sales		Trend val
1990	170	$\bar{x}_1 = 232.25$	$224.5 - 15.5 = 209$
1991	231		$232.25 - 7.75 = 224.5$
1992	261		centre = 232.25
1993	267		$232.25 + 7.75 = 240$
1994	278		$240 + 15.5 = 255.5$
1995	302		
1996	299	$\bar{x}_2 = 309.75$	$302 - 15.5 = 286.5$
1997	298		$309.75 - 7.75 = 302$
1998	340		centre = 309.75
			$309.75 + 7.75 = 317.5$
			$317.5 + 15.5 = 333$

$$\text{annual sales} = \frac{\bar{x}_2 - \bar{x}_1}{5} = \frac{309.75 - 232.25}{5} = 15.5$$

$$\text{half yearly increment} = \frac{15.5}{2} = 7.75$$

for 1999 the sales is $333 + 15.5 = 348.5$



Method of least Squares

$$y = a + bx$$

$$n = \text{odd}$$

$$x = \frac{t - \text{middle year}}{\text{Interval}}$$

$$a = \frac{\sum y}{n} \quad b = \frac{\sum xy}{\sum x^2}$$

Interval

$$n = \text{even}$$

$$x = \frac{t - (\text{AM of 2 middle year})}{1/2 [\text{Interval}]}$$

$$1/2 [\text{Interval}]$$

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

1. The decision making body of a fertilizer firm ^{producing} wants to predict future sales trend ^{of 2006 & 2008} based on the analysis of its past sales pattern. The sales of the firm for the last 7 years for this purpose are given.

Year	Sales	$x = t - 2001$	x^2	xy	$y = 87 + 4.18x$
1998	70	-3	9	-210	74.46
1999	75	-2	4	-150	78.64
2000	90	-1	1	-90	82.82
2001	98	0	0	0	87
2002	85	1	1	85	91.178
2003	91	2	4	182	95.356
2004	100	3	9	300	99.534

$$\sum y = 609$$

$$\sum x^2 = 28$$

$$\sum xy = 117$$

$$a = \frac{\sum y}{n} = \frac{609}{7} = 87$$

$$b = \frac{\sum xy}{\sum x^2} = \frac{117}{28} = 4.178$$

$$y = a + bx$$

$$= 87 + 4.178x$$

$$2006 \rightarrow x = 2006 - 2001 = 5$$

2005

$$y = 87 + 4.18(5) = 107.89$$

$$2008 \rightarrow x = 2008 - 2001 = 7$$

$$y = 116.246$$

2. Fit a linear trend to the following data by the least square method. Estimate the production for the year 1999.

$$y = a + b \cdot x$$

year	sales	$x = \frac{t - 1994}{2}$	x^2	xy	y
1990	18	-2	4	-36	20.6
1992	21	-1	1	-21	20.8
1994	23	0	0	0	21
1996	27	1	1	27	21.2
1998	16	2	4	32	21.4
$\Sigma y = 105$			10	2	

$$a = \frac{\Sigma y}{n} = \frac{105}{5} = 21$$

$$b = \frac{\Sigma xy}{\Sigma x^2} = \frac{2}{10} = 0.2$$

for 1999 $\bar{x} = \frac{1999 - 1994}{2}$
 $= 2.5$

$$x = 2.5$$

$$y = 21 + 0.2(2.5) = 21.5$$

3. A district sales office of a leading fast moving consumer goods company has been tracking the yearly sales of one of its flagship products, the company's management team is looking to make and inform strategic decision regarding inventory marketing and resource allocation. To support this they want to understand the long term trend in the sales performance. the data is given as

year	sales	$x = \frac{t - 2003.5}{1/2(1)}$	x^2	xy	$y = 46.75 + 0.881x$
2000	40	-3.5	12.25	-140	40.583
2001	45	-2.5	6.25	-112.5	42.345
2002	46	-1.5	2.25	-69	44.107
2003	42	-0.5	0.25	-21	45.869
2004	47	0.5	0.25	23.5	47.631
2005	50	1.5	2.25	76.5	49.393
2006	46	2.5	6.25	119	51.155
2007	58	3.5	12.25	208	52.917
	374		48	148	

AM of 2 mid = $\frac{2003 + 2004}{2} = 2003.5$

$$a = \frac{\sum y}{n} = \frac{374}{8} = 46.75$$

$$y = a + bx$$

$$= 46.75 + 0.881x$$

$$b = \frac{\sum xy}{\sum x^2} = \frac{148}{168} = 0.881$$

4. Fit a straight line trend for the following data

Year	Electricity generation	$x = \frac{t - 1953.5}{1/2(1)}$	x^2	xy	$y = 121.33 + 4.686x$
1951	101	-2.5	25	-505	97.902
1952	107	-2	4	-321	107.272
1953	115	-1	1	-115	116.641
1954	121	0	0	0	121.33
1955	136	1	1	136	126.016
1956	148	2	4	296	135.388
	728		70	328	144.76

$$a = \frac{\sum y}{n} = \frac{728}{6} = 121.33 \quad b = \frac{\sum xy}{\sum x^2} = \frac{164}{17.5} = 9.371$$

$$a = \frac{\sum y}{n} = \frac{728}{6} = 121.33 \quad b = \frac{\sum xy}{\sum x^2} = \frac{328}{70} = 4.686$$

$$121.33 + 4.6857x$$

Method of moving average

Obtaining a series of moving average. The moving average of period m is series of successive average of m overlapping values at a time starting with 1st and m th value and so on.

$$1^{st} MA = \frac{y_1 + y_2 + \dots + y_m}{m}$$

$$2^{nd} MA = \frac{y_2 + y_3 + \dots + y_{m+1}}{m}$$

$$3^{rd} MA = \frac{y_3 + y_4 + \dots + y_{m+2}}{m}$$

$$3 \text{ yearly MA} = \frac{a+b+c}{3}, \frac{b+c+d}{3}$$

$$4 \text{ yearly MA} = \frac{a+b+c+d}{4}, \frac{b+c+d+e}{4}$$

Method of mo

1. Calculate 3 yearly & 5 yearly moving average for the following data -

$$242 + 250 + 252 = \frac{744}{3}$$

Year	Sales	Total	3 yearly MA	total	5 yearly
1990	242	-	-	-	-
1991	250	242+250+252	$\frac{744}{3} = 248$	248 -	-
1992	252	250+252+249	250.33	242+250+252 + 249+253	$\frac{1246 + 249.2}{5} =$
1993	249	252+249+253	251.33	250+252+249 + 253+255	251.8
1994	253	249+253+255	252.33	252+249+253 + 255+251	252
1995	255	253+255+251	253	249+253+255 + 251+257	253
1996	251	255+251+257	254.33	253+255+251 + 257+260	255.2
1997	257	251+257+260	256	255+251+257 + 260+265	257.6
1998	260	257+260+265	260.67	257+260+265 + 262+265	259
1999	265	260+265+262	262.33	-	-
2000	262	-	-	-	-

Calculate trend value

$$251 + 257 + 260 + 265 + 262$$

Year	Sugar production	4 year total	4 year MA	4 year centroid
1971	37.4	-	-	-
1972	31.1	-	-	-
1973	38.7	37.4+31.1+38.7+39.5	$\frac{146.7}{4} = 36.675$	37.99
1974	39.5	31.1+38.7+39.5+47.9	39.3	40.7375
1975	47.9	38.7+39.5+47.9+42.6	42.175 44.6	43.387
1976	42.6	39.5+47.9+42.6+48.4	35.68 50.875	47.737
1977	48.4	47.9+42.6+48.4+64.6	53.5	52.19
1978	64.6	42.6+48.4+64.6+58.4	52.5	53
1979	58.4	48.4+64.6+58.4+38.6	53.35	52.925
1980	38.6	64.6+58.4+38.6+51.4	58.2	55.775
1981	51.4	58.4+38.6+51.4+84.4	-	-
1982	84.4	-	-	-

Ratio to Trend:

- Steps to calculate Ratio to trend method:
- Step 1: Obtain the trend values by the method of least squares for each time period by fitting a straight line
 - Step 2: Divide each original value by corresponding trend value & multiply it by 100
 - Step 3: Find the average of ratio to trend values for each season for any number of years.
If the seasonal periods are quarters sum of seasonal indices for multiplicative model should be 400, if the periods are in months it should be 1200.

4.

year	Summer	monsoon	winter	Spring	Aug.
2019	70	52	22	31	43.75
2020	101	64	44	45	63.5
2021	120	75	30	49	68.5
2022	135	82	34	50	75.25

$$y = a + bx$$

Original Data (Original Set)

year	y	$x = \frac{t - 2020.5}{1/2(1)}$	x^2	xy	$y = 62.75 + 4.975x$
2019	43.75	-3	9	-131.25	47.825
2020	63.5	-1	1	-63.5	57.775
2021	68.5	1	1	68.5	67.725
2022	75.25	3	9	225.75	77.675
	251		20	99.5	

$$a = \frac{\sum y}{n} = \frac{251}{4} = 62.75$$

$$\text{annual increment} = 4.975$$

$$b = \frac{\sum xy}{\sum x^2} = \frac{99.5}{20} = 4.975$$

$$\text{quarterly increment} = \frac{4.975}{4} = 1.244$$

$$\text{Half quarterly increment (HQR)} = \frac{1.24}{2} = 0.62$$

Year	I	II	TV
2019			47.82
2020			57.77
2021			67.42
2022			77.67

Quarterly Trend

	II - QI TV - QI	TV - HQI	TV	TV + HQI	III + QI
Year	I	II		III	IV
2019	45.96	47.2	47.82	48.44	49.68
2020	55.91	57.15	57.77	58.39	59.63
2021	65.84	67.08	67.42	68.34	69.58
2022	75.81	77.05	77.67	78.29	79.53

Calculation of trend value and seasonal variation index.

$$\frac{\text{Original Qrt}}{\text{Qrt trend val}} \times 100$$

Year	Summer	Monsoon Qrt	Winter	Spring
2019	$\frac{70}{45.96} \times 100 = 152.31$	110.17	45.417	62.399
2020	180.65	111.986	75.355	75.465
2021	182.26	111.807	43.898	70.4225
2022	178.07	106.424	43.428	62.869